

QuadSync Tech Solutions

Synchronizing Innovations With Success

Web Development Internship (Syllabus)

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1.	HTML	4-5 Hrs	ii
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MSME
MICRO, SMALL & MEDIUM ENTERPRISES

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HTML:

Chapter No	Topic	Days
Chapter 1	Introduction to Web Development, How the Web Works, HTML Structure (doctype, html, head, body), Building Your First Webpage	Day 1
Chapter 2	Text Formatting, Headings, Paragraphs, Links, Images, Ordered & Unordered Lists	Day 2
Chapter 3	Tables, Forms (input types, labels, buttons), Semantic Tags (header, section, footer)	Day 3
Chapter 4	HTML Practice: Create a Registration Form + Multi-section Webpage	Day 4
Chapter 5	Forms Deep-dive, Accessibility Basics (alt text, labels, ARIA attributes), HTML Recap & Mini Quiz	Day 5

CSS:

Chapter No	Topic	Days
Chapter 6	Introduction to CSS, Types of CSS (Inline, Internal, External), Selectors (class, id), Colors & Fonts	Day 6
Chapter 7	Box Model (Margin, Padding, Border), Width & Height	Day 7
Chapter 8	Display Properties (block, inline, inline-block), Positioning (relative, absolute, fixed, sticky)	Day 8
Chapter 9	Flexbox (Container, Alignment, Layout), Navbar Design	Day 9
Chapter 10	CSS Grid Layout Basics (grid-template-columns, gap, areas)	Day 10
Chapter 11	Responsive Design (Media Queries), Mobile-First Layout	Day 11
Chapter 12	CSS Practice Project: Build a Responsive Landing Page	Day 12

Tailwind CSS:

Chapter No	Topic	Days
Chapter 13	Introduction to Tailwind CSS, Setup & Configuration, Utility-first Concept	Day 13
Chapter 14	Tailwind Spacing, Typography & Colour Utility Classes	Day 14
Chapter 15	Tailwind Flexbox & Grid Utilities	Day 15
Chapter 16	Tailwind Responsive Design (Breakpoints: sm, md, lg, xl)	Day 16
Chapter 17	Building Components with Tailwind: Buttons, Cards, Forms	Day 17
Chapter 18	Tailwind Practice: Build a Responsive Navbar & Hero Section	Day 18
Chapter 19	Combining Custom CSS with Tailwind, Best Practices	Day 19
Chapter 20	CSS & Tailwind Project: Style a Complete Portfolio Landing Page	Day 20

JavaScript:

Chapter No	Topic	Days
Chapter 21	JavaScript Introduction, Variables (let , const), Data Types, Operators	Day 21
Chapter 22	Conditional Statements (if-else, nested), Loops (for, while)	Day 22
Chapter 23	Functions (Parameters, Return Values), Basic Programs (Calculator Logic)	Day 23
Chapter 24	Arrays & Array Methods (map, filter, forEach, reduce)	Day 24
Chapter 25	Objects, Object Methods & JSON Basics	Day 25
Chapter 26	DOM Manipulation, Selecting & Updating Elements	Day 26
Chapter 27	Events (click, input, submit), Event Listeners	Day 27
Chapter 28	Form Validation with JavaScript	Day 28
Chapter 29	JavaScript Project: Build a To-Do App / Calculator	Day 29
Chapter 30	Project Completion, Testing & Debugging	Day 30

React JS:

Chapter No	Topic	Days
Chapter 31	Introduction to React, Project Setup (Vite), JSX Basics	Day 31
Chapter 32	Functional Components & Props	Day 32
Chapter 33	State Management with useState	Day 33
Chapter 34	Handling Events & Building Forms in React	Day 34
Chapter 35	useEffect Hook, Side Effects & Component Lifecycle	Day 35
Chapter 36	Conditional Rendering, Rendering Lists & Keys	Day 36
Chapter 37	React Router: Navigation & Multi-page Apps	Day 37
Chapter 38	API Integration with fetch/axios, Displaying Dynamic Data	Day 38
Chapter 39	React Project: Build a Weather App / Product Listing App	Day 39
Chapter 40	Project Completion & Styling with Tailwind CSS	Day 40

Express, Node & MongoDB:

Chapter No	Topic	Days
Chapter 41	Introduction to Node.js, npm, Modules	Day 41
Chapter 42	Introduction to Express.js, Setting up a Server, Routing	Day 42
Chapter 43	Middleware, Request & Response Handling	Day 43
Chapter 44	REST API Basics, Building CRUD APIs	Day 44
Chapter 45	Introduction to MongoDB, Databases & Collections	Day 45
Chapter 46	Mongoose: Schemas & Models	Day 46
Chapter 47	Connecting Express with MongoDB: Full CRUD Operations	Day 47
Chapter 48	Authentication Basics (JWT & bcrypt Introduction)	Day 48
Chapter 49	Backend Project: Build a REST API for a To-Do / Blog App	Day 49
Chapter 50	Testing APIs with Postman, Error Handling	Day 50

Git, GitHub & Final Project:

Chapter No	Topic	Days
Chapter 51	Git Basics: init, add, commit, status, log	Day 51
Chapter 52	Branching & Merging, .gitignore	Day 52
Chapter 53	GitHub: Creating Repositories, Push/Pull, Remotes	Day 53
Chapter 54	Collaboration: Pull Requests & Resolving Merge Conflicts	Day 54
Chapter 55	Final Project Planning: Designing a Full-Stack (MERN) App	Day 55
Chapter 56	Final Project: Frontend Development	Day 56
Chapter 57	Final Project: Backend Development & API Integration	Day 57
Chapter 58	Final Project: Connecting Frontend & Backend	Day 58
Chapter 59	Deployment (Vercel/Netlify for Frontend, Render/Railway for Backend)	Day 59
Chapter 60	Final Project Testing, Presentation & Portfolio Building	Day 60